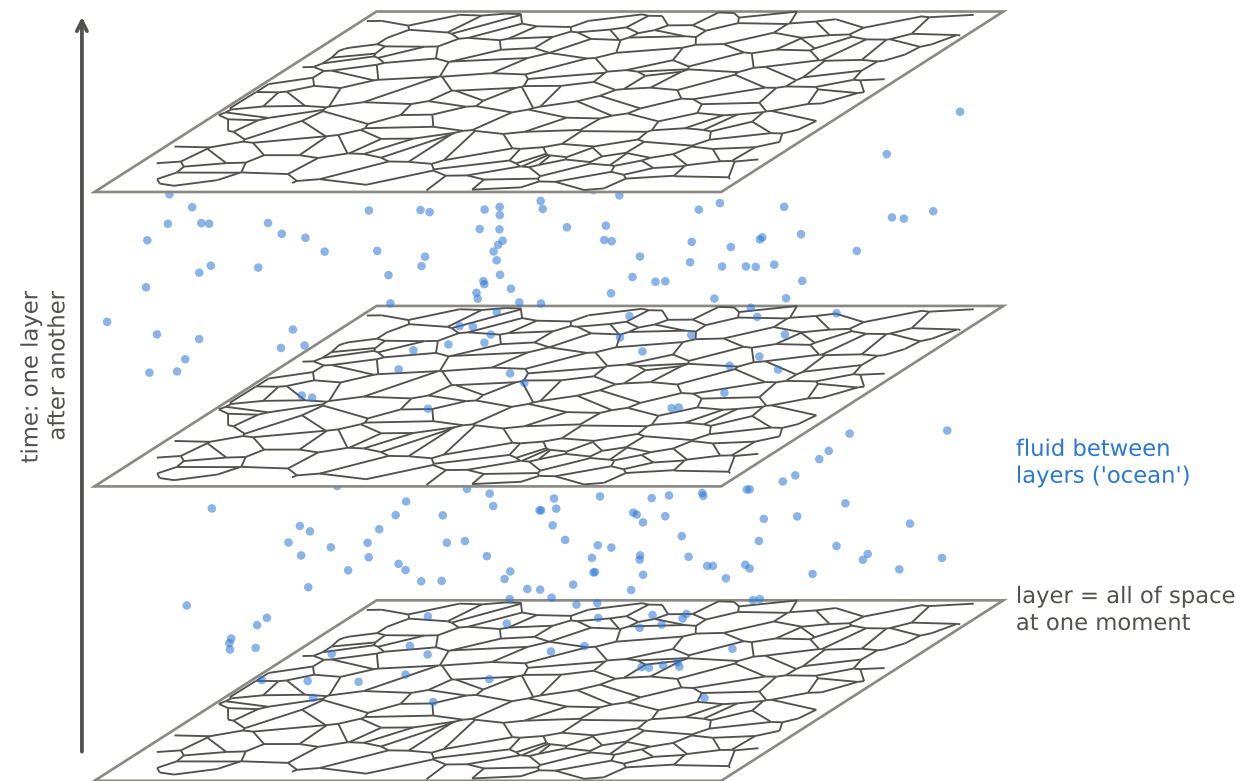


Figure 1. Anatomy of the grid



WHAT THE PIECES ARE

Links: carry tension, like the strings of a stretched net. Their pull is what we feel as gravity (Fig. 2-3).

Cells: random, not a crystal, so no direction is special (a regular grid would show its axes; we measured $\sim 20\%$ for a cube lattice).

Layers: one per moment, like the pages of a flip-book. Relativity's 'preferred slicing' in Khronon theory is this stack.

Fluid: fills the gaps between pages. It is pulled by gravity but does not collide or shine: the dark matter of the CMB, clusters, lensing.

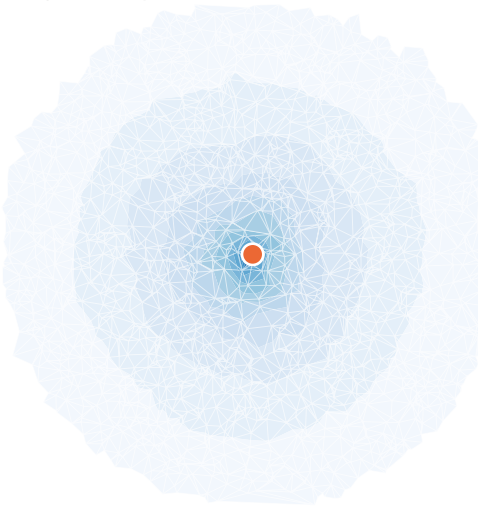
Cosmic tension: the net's overall stretch is the dark energy (Fig. 4).

picture (layout); cell geometry is a real random mosaic

Figure 2. Gravity is the grid's tension, shared out link by link

(a) A mass pulls on the net

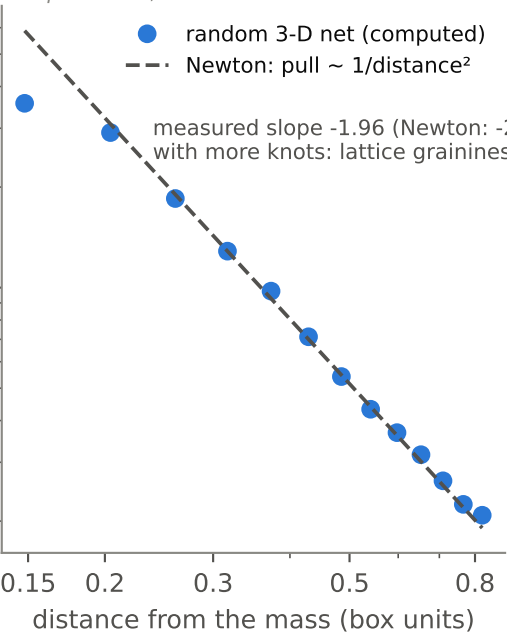
computed: equilibrium of a 2-D random network



a random net, tension only; the mass pulls one knot.
darker = pulled further. Every knot shares the load
with its neighbours, so the dip spreads out smoothly.

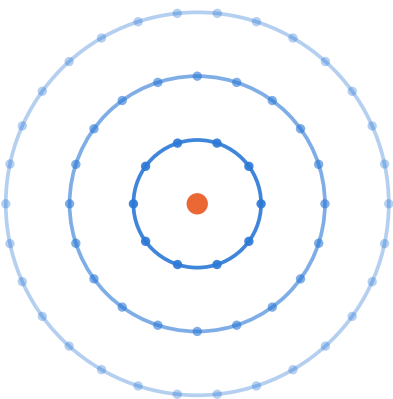
(b) The pull falls as $1/r^2$

computed: 24,000-knot random 3-D network



(c) Why $1/r^2$

picture



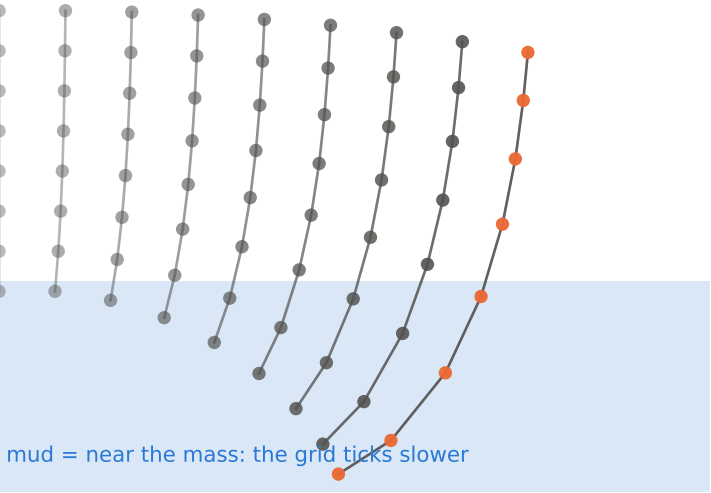
the same pull is shared by every link crossing
a shell; a shell twice as far has 4x as many,
so each carries 1/4: the inverse-square law.

Figure 3. What gravity actually is: the grid ticks slower near mass, and paths bend toward slow ticking

(a) Falling = turning toward slower ticks

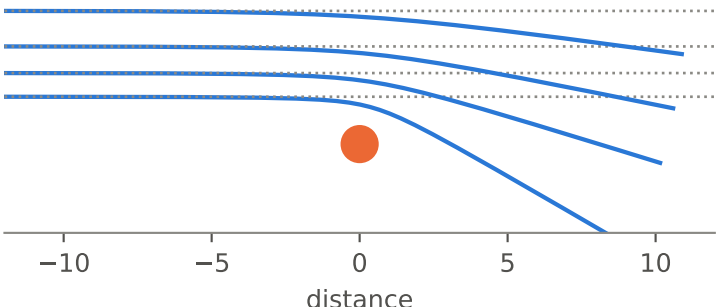
picture (computed toy: step length = local tick rate)

a marching band whose lower ranks reach mud
each row turns toward the slow side:
no rope pulls it; the path itself bends



(b) Light takes the same bent path

computed: ray tracing through the tick-rate field (exaggerated)



dotted: straight lines. Rays passing closer bend more.
The Sun bends starlight by 1.75" (measured 1919 onward);
here the mass is exaggerated $\sim 10^5$ times.

Figure 4. Dark energy is the grid's own tension, and it 'gives' when the stretching speeds up

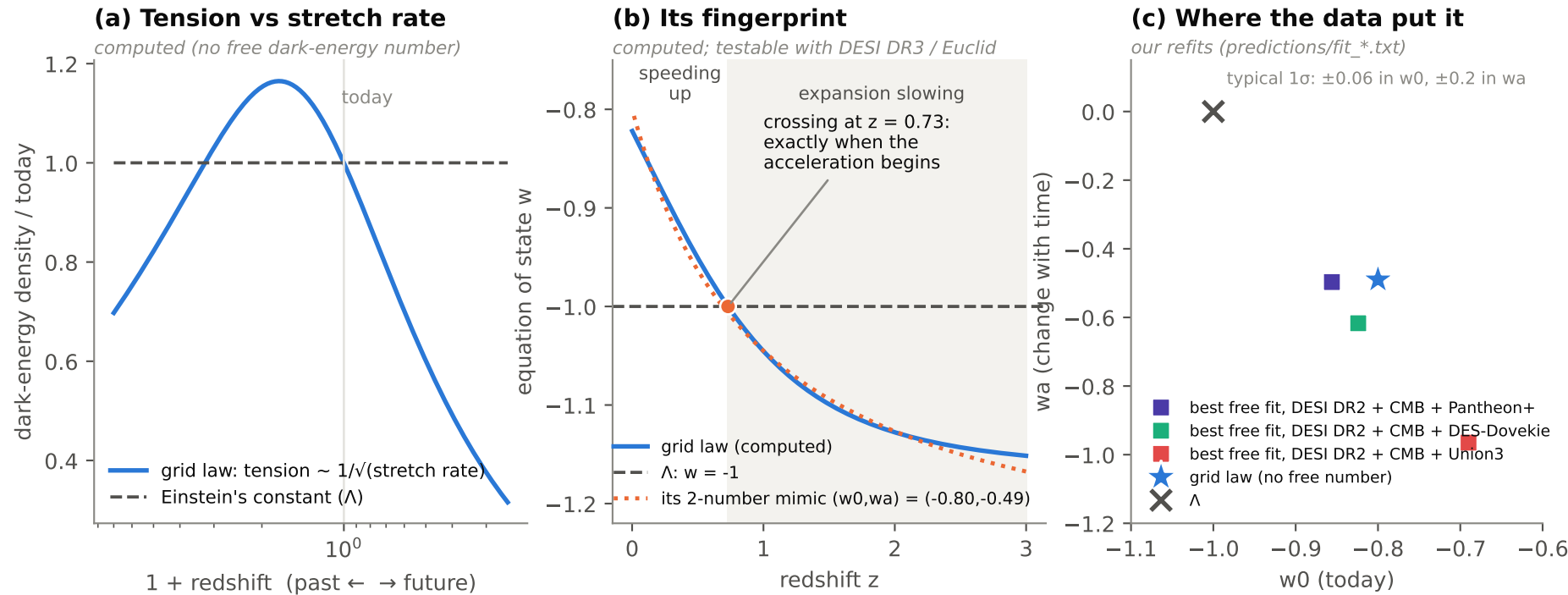
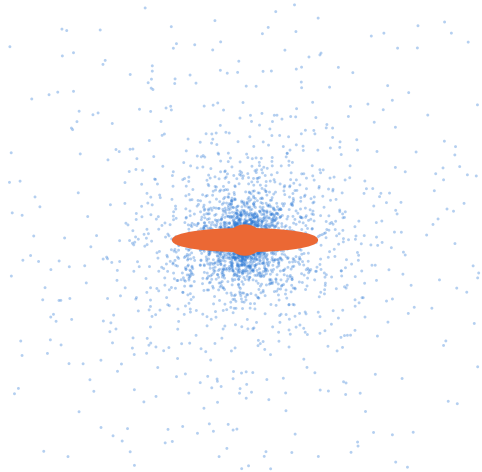


Figure 5. The fluid between the layers: pulled by gravity, invisible, and it passes through itself

(a) A galaxy sits in a pool of fluid

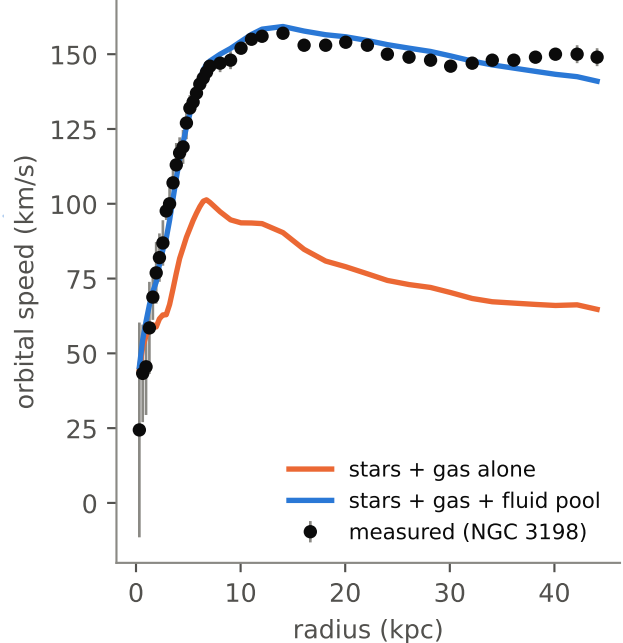
picture (profile drawn from a CDM-like halo)



orange: stars and gas. blue: the fluid it has gathered (5-6x more mass). The pool holds the outer stars in orbit (flat rotation curves).

(b) Why we need the fluid

computed: SPARC data, Newtonian + cored halo fit



(c) Two clusters after a collision

picture of the Bullet Cluster observation

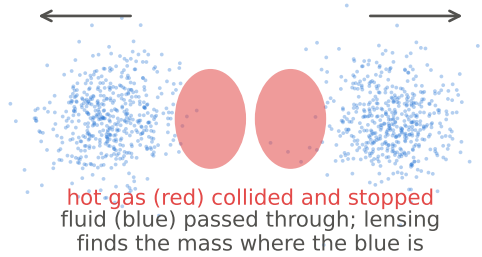


Figure 6. The history of the universe in the grid picture, with what has been tested

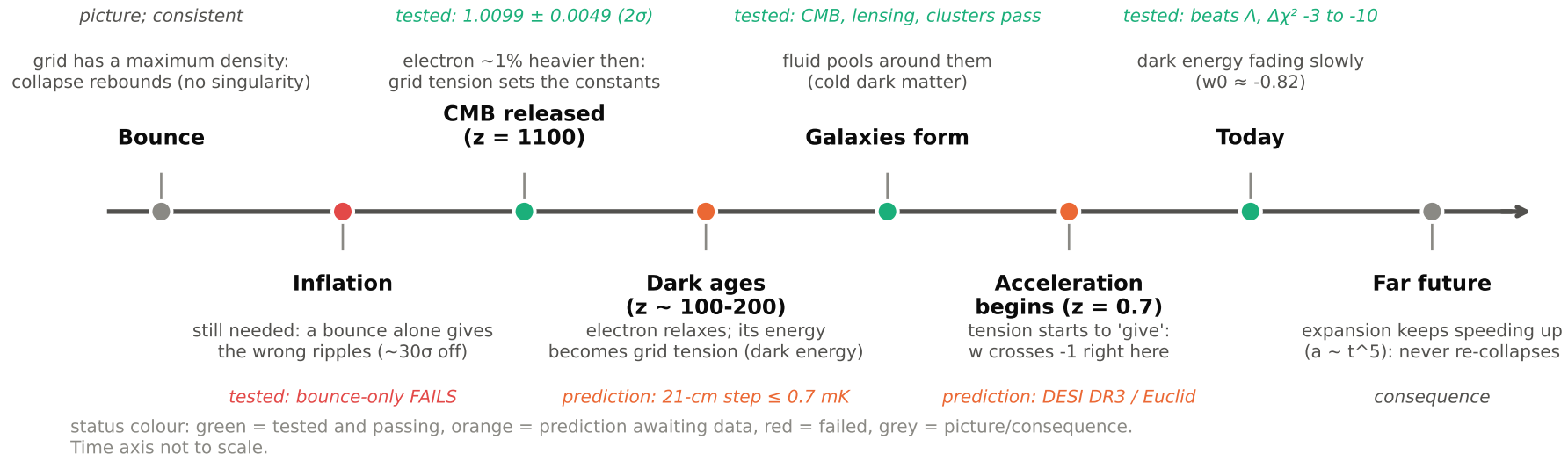


Figure 7. Matter as patterns in the grid: the electron's two handed halves live on two layers

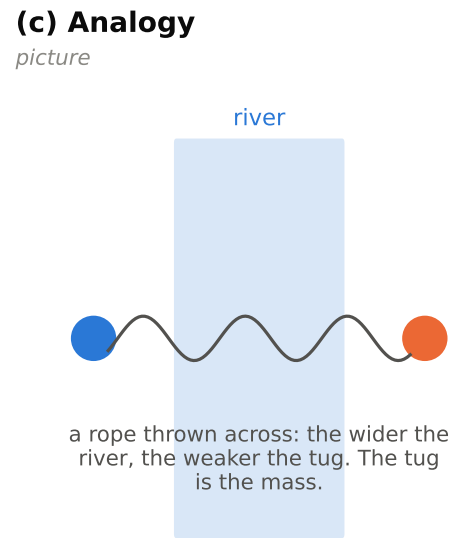
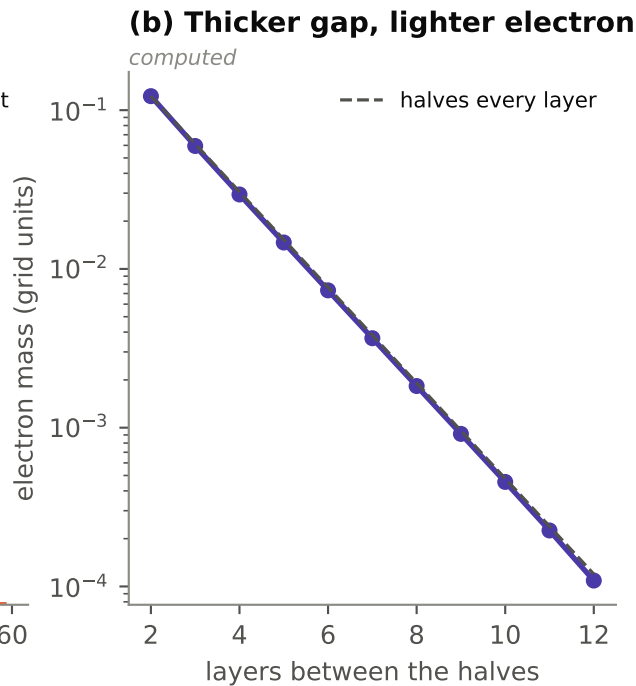
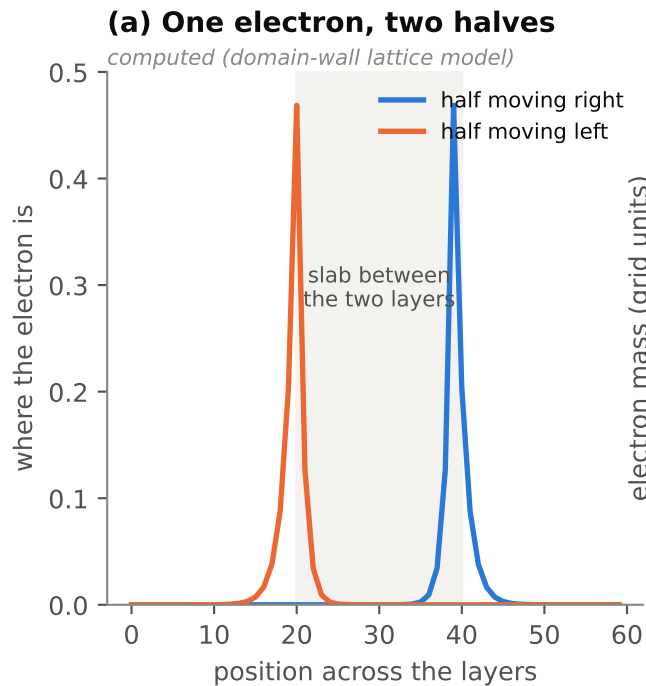
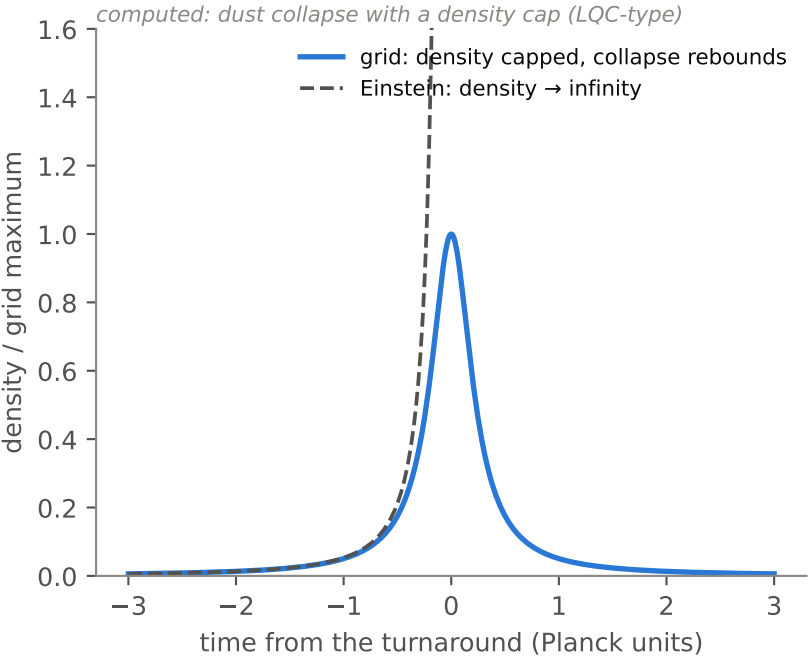


Figure 8. Black holes: the grid cannot be compressed without limit

(a) The core of a collapsing star



(b) Analogy

picture

a full lift: people cannot squeeze any closer,
so the push turns into a rebound

The diagram shows a 4x5 grid of circles. The circles in the first and third columns are orange, while the circles in the second and fourth columns are blue. This creates a checkerboard pattern of colors across the grid.

result: a 'Planck star' at the centre, hidden by the horizon;
no observable difference for $>1e26$ years

Figure 9. Was the electron heavier when the CMB was released?

